7. Find the separate equation of the following pair of straight lines
   (i) $3x^2 + 2xy - y^2 = 0$
   (ii) $6(x-1)^2 + 5(x-1)(y-2) - 4(y-2)^2 = 0$
   (iii) $2x^2 - xy - 3y^2 - 6x + 19y - 20 = 0$
8. The slope of one of the straight lines $ax^2 + 2hxy + by^2 = 0$ is twice that of the other, show that $8h^2 = 9ab$.
9. The slope of one of the straight lines $ax^2 + 2hxy + by^2 = 0$ is three times the other, show that $3h^2 = 4ab$.
10. A ΔOPQ is formed by the pair of straight lines $x^2 - 4xy + y^2 = 0$ and the line PQ. The equation of PQ is $x + y - 2 = 0$. Find the equation of the median of the triangle ΔOPQ drawn from the origin O.
11. Find p and q, if the following equation represents a pair of perpendicular lines

$$6x^2 + 5xy - py^2 + 7x + qy - 5 = 0$$

12. Find the value of k, if the following equation represents a pair of straight lines. Further, find whether these lines are parallel or intersecting, $12x^2 + 7xy - 12y^2 - x + 7y + k = 0$
13. For what value of k does the equation $12x^2 + 2kxy + 2y^2 + 11x - 5y + 2 = 0$ represent two straight lines.
14. Show that the equation $9x^2 - 24xy + 16y^2 - 12x + 16y - 12 = 0$ represents a pair of parallel lines. Find the distance between them.
15. Show that the equation $4x^2 + 4xy + y^2 - 6x - 3y - 4 = 0$ represents a pair of parallel lines. Find the distance between them.
16. Prove that one of the straight lines given by $ax^2 + 2hxy + by^2 = 0$ will bisect the angle between the co-ordinate axes if $(a+b)^2 = 4h^2$
17. If the pair of straight lines $x^2 - 2kxy - y^2 = 0$ bisect the angle between the pair of straight lines $x^2 - 2lxy - y^2 = 0$, Show that the later pair also bisects the angle between the former.
18. Prove that the straight lines joining the origin to the points of intersection of $3x^2 + 5xy - 3y^2 + 2x + 3y = 0$ and $3x - 2y - 1 = 0$ are at right angles.



Exercise - 6.5



Choose the correct or more suitable answer

1. The equation of the locus of the point whose distance from y-axis is half the distance from origin is
   (1) $x^2 + 3y^2 = 0$ (2) $x^2 - 3y^2 = 0$ (3) $3x^2 + y^2 = 0$ (4) $3x^2 - y^2 = 0$
2. Which of the following equation is the locus of $(at^2, 2at)$
   (1) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (2) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (3) $x^2 + y^2 = a^2$ (4) $y^2 = 4ax$
3. Which of the following point lie on the locus of $3x^2 + 3y^2 - 8x - 12y + 17 = 0$
   (1) $(0,0)$ (2) $(-2,3)$ (3) $(1,2)$ (4) $(0,-1)$
4. If the point (8,-5) lies on the locus $\frac{x^2}{16} - \frac{y^2}{25} = k$, then the value of k is
   (1) 0 (2) 1 (3) 2 (4) 3